

Experiment: Determination of Sulphide Content of Tannery Effluent

Iodometric Method — Reference standard: APHA Standard Methods 4500-S²⁻ F

Aim

To measure how much sulphide is present in a tannery effluent stream, using the standard iodometric titration, and to compare the result with the discharge limit.

Principle (in simple terms)

Sulphide cannot be titrated directly, because as soon as acid is added it escapes as toxic H₂S gas. So we use a back-titration — like counting change from a shopkeeper:

- We give the sulphide a known, extra-large amount of iodine to react with.
- The sulphide “uses up” part of that iodine.
- We then measure how much iodine is left over, by titrating it with sodium thiosulphate.
- Iodine used by the sulphide = iodine added – iodine left over. From that we calculate the sulphide.

The reactions are:



1 mL of 0.025 N iodine is equal to 0.400 mg of sulphide.

Why the sample must be “fixed” with zinc acetate

Sulphide disappears within minutes of leaving the tank — it oxidises in air and escapes as H₂S gas. To stop this, we add zinc acetate solution to the sample bottle at the moment of collection. The zinc ion locks the sulphide up as a solid, zinc sulphide:



Trapped as ZnS, the sulphide can no longer escape or oxidise, so it survives the trip to the lab. In the titration, the acid simply releases it again — right in the flask where the iodine is waiting. (This is the milky flocc you will see in the sample.)

Group Allocation — Four Groups

Each group tests a different tannery stream for sulphide. Pooling the four results shows where sulphide concentrates — highest in the lime (beamhouse) stream, lowest in the treated outlet.

Group	Assigned stream	Members
Group 1	Soak liquor (soaking)	_____
Group 2	Lime liquor (liming / unhairing) — expect highest	_____
Group 3	Composite raw effluent (ETP inlet)	_____
Group 4	Treated effluent (ETP outlet) — expect lowest	_____

Apparatus

- Burette (25 mL), pipettes, measuring cylinder
- Glass-stoppered conical (iodine) flasks
- Fume hood
- Sample bottles (for fixed samples)

Reagents (and what each one does)

- **Standard iodine solution, 0.025 N** — the known excess that reacts with the sulphide
- **Standard sodium thiosulphate, 0.025 N** — used to measure the leftover iodine
- **Starch indicator** — shows the exact end-point (blue → colourless)
- **Dilute acid (6 N HCl)** — releases the sulphide from the ZnS floc
- **Zinc acetate (2 N) + sodium hydroxide (6 N)** — fix / preserve the sulphide at sampling

Note: the iodine and thiosulphate must be standardised (their exact normality known), or the result will be wrong.

Sample Collection & Fixation

At the sampling point: add 2 N zinc acetate to the sample until a pale ZnS floc forms, then raise the pH above 9 with a little 6 N NaOH. Fill the bottle to the brim (no air bubbles), keep it cold and dark, and analyse as soon as possible.

Procedure

1. Work in a fume hood throughout (H₂S is toxic).
2. Into a flask, first add a measured, known volume of standard iodine (for example 25.0 mL — write it down) and a few mL of dilute acid. Putting the iodine in first means it captures the sulphide the instant it is released.
3. Add your fixed sample (with the milky floc) to the iodine–acid mixture. The solution must stay yellow-brown — that shows iodine is still in excess. If it goes clear, add more measured iodine and note the extra.
4. Fill the burette with thiosulphate and titrate the leftover iodine — the yellow-brown colour slowly fades.
5. When the colour is pale straw, add a few drops of starch (it turns blue-black). Continue drop by drop until the blue just disappears — that is the end-point. Record the thiosulphate used. Add starch late, or the end-point becomes fuzzy.

Observations

A. Your own group's data

Quantity	Value
Volume of iodine added, A (mL)	
Normality of iodine, B	

Quantity	Value
Thiosulphate used, C (mL)	
Normality of thiosulphate, D	
Blank titre (mL)	
Sample volume, V (mL)	

B. Pooled class results (copy every group's result)

Stream	Sulphide (mg/L)
Soak liquor	
Lime liquor (beamhouse)	
Composite raw effluent	
Treated outlet	

Calculation

$$\text{Sulphide (mg S}^{2-}\text{/L)} = [(A \times B) - (C \times D)] \times 16\,000 / V$$

where A = mL iodine, B = normality of iodine, C = mL thiosulphate, D = normality of thiosulphate, V = sample volume (mL).

Precautions

- Fix the sample with zinc acetate immediately at sampling — unpreserved sulphide is lost within minutes.
- Acidify only in a fume hood (H₂S gas).
- Keep iodine in excess (yellow-brown) before back-titrating, or the result reads low.
- Use standardised iodine and thiosulphate; add starch only near the end-point.

Result & Interpretation

Report the sulphide as mg/L (blank-corrected average). Compare with the Bangladesh ECR 2023 inland-discharge limit of 2.0 mg/L. Raw lime/beamhouse liquor (often 100–300 mg/L) will be far above the limit, showing why sulphide must be oxidised in primary treatment before the streams are mixed.